

deal with tons of data. If this data is refined to a useful form, only then is it of some real use.

As an example, we can see the transformation televisions have undergone, in appearance. We once had picture tube based TVs. Today, we have LEDs, OLEDs, LCD based TVs, curved TVs, Internet enabled TVs, and so on. Such transformation is also quite evident in the digital world.

In a hospital, several patients may be diagnosed with cancer, each year. The patient data generated is voluminous, including treatment methods, diverse drug therapies, patient responses, genetic histories, etc. But such vast pools of information, i.e., Big Data, would serve no useful purpose without proper analysis. So DT, coupled with Big Data and open source applications, can create a more patient-focused and effective treatment – one that might have higher recovery rates.

Big Data combines structured data with unstructured data to give us new business insights that we've never had before. Structured data may be traditional spreadsheets, your customer list, information about your products and business processes, etc. Unstructured data may include Google Trends data, feeds from IoT sensors, etc. When a layer of unstructured data is placed on top of structured data and analysed, that's where the magic happens.

Let's look into a typical business situation. Let's suppose a century old car-making company asks its data team to use Big Data concepts to find an efficient way to make safe sales forecasts. In the past, the team would look at the number of products it had sold in the previous month, as well as the number of cars it had sold a year ago and use that data to make a safe forecast. But now the Big Data teams use sentiment analysis on Twitter and look at what people are saying about its products and brand. They also look at Google Trends to see which similar products and brands are being searched the most. Then they

correlate such data from the preceding few months with the actual current sales figures to check if the former was predictive – i.e., had Google Trends over the past few months actually predicted the firm's current sales figures?

In the case of the car company, while making sales forecasts, the team used structured data (how many cars sold last month, a year ago, etc) and layers of unstructured data (sentiment analysis from Twitter and Google Trends) and it resulted in a smart forecast. Thus, Big Data is today becoming more effective in business situations like sales planning, promotions, market campaigns, etc.

Open source is the key to DT

Open source, nowadays, clearly dominates domains like Big Data, mobile and cloud platforms. Once open source becomes a key component that delivers a good financial performance, the momentum is unstoppable. Open source (often coupled with the cloud) is giving Big Data based companies like Google, Facebook and other Web giants flexibility to innovate faster.

Big Data companies are using DT to understand their processes, so that they can employ technologies like IoT, Big Data analytics, AI, etc, better. The journey of enterprises migrating from old digital infrastructure to new platforms is an exciting trend in the open source environment.

Organisations are relying on data warehouses and business intelligence applications to help make important data driven business decisions. Different types of data, such as audio, video or unstructured data, is organised in formats to help identify it for making future decisions.

Open source tools used in DT

Several open source tools are becoming popular for dealing with Big Data and DT. Some of them are listed below.

- **Hadoop** is known for the ability to process extremely large data

volumes in both structured and unstructured formats, reliably placing Big Data to nodes in the group and making it available locally on the processing machine.

- **MapReduce** happens to be a crucial component of Hadoop. It works rapidly to process vast amounts of data in parallel on large clusters of computer nodes. It was originally developed by Google.
- **Storm** is different from other tools with its distributed, real-time, fault-tolerant processing system, unlike the batch processing of Hadoop. It is fast and highly scalable. It is now owned by Twitter.
- **Apache Cassandra** is used by many organisations with large, active data sets, including Netflix, Twitter, Urban Airship, Cisco and Digg. Originally developed by Facebook, it is now managed by the Apache Foundation.
- **Kaggle** is the world's largest Big Data community. It helps organisations and researchers to post their data and statistics. It is an open source Big Data tool that allows programmers to analyse large data sets on Hadoop. It helps with querying and managing large data sets really fast.

DT: A new innovation

DT is the result of IT innovation. It is driven by well-planned business strategies, with the goal of inventing new business models. Today, any organisation can undergo business transformation because of three main business-focused essentials — intelligence, the ability to decide more quickly and a customer-centric outlook.

DT, which includes establishing Big Data analytics capabilities, poses considerable challenges for traditional manufacturing organisations, such as car companies. The successful introduction of Big Data analytics often requires substantial organisational transformation including new